

Project I – Web-Based Application Project

Project Documentation Requirements

Each group must submit a project report that explains the problem, development process, and the implemented web-based solution.

1. Cover Page

Include:

- Project Title
- Assigned Challenge
- Group Members (Names and Student Numbers)

2. Introduction

Provide a brief overview of:

- The problem being addressed
- The importance of the proposed solution

3. Problem Statement and Objectives

Clearly explain:

- The assigned problem or challenge
- Who is affected by the problem
- The objectives of the web app
- Expected outcomes of the solution

4. Methodology (Crystal Yellow)

Describe how the Crystal (Yellow) Agile methodology was applied.

Include:

- Team roles and collaboration
- Iterative development approach
- Task planning and progress tracking

5. System Design

Explain how the system was designed.

Include:

- System architecture overview
- Database design (ERD and relationships)
- User interface design or wireframes

6. Technology Stack

Describe the technologies used and support the choice:

- **Frontend**
- **Backend**
- **Database**

7. System Implementation

Explain how the system was developed.

Include:

- Application structure (routes, models, templates)
- Key system features
- Screenshots of important parts of the application where applicable (e.g., dashboard, forms, admin panel)

8. Security and Data Management

Briefly explain how the following is achieved in the developed solution:

- User authentication and role-based access control
- Password hashing
- Input validation and protection against common vulnerabilities
- Consideration of POPIA compliance

9. Testing

Describe how the web app was tested.

Include examples of:

- Functional testing and non-functional testing
- Test cases and results

10. Deployment

Explain how the system was deployed.

Include:

- Link to the deployed application
- Hosting platform used

11. Source Code Repository

Provide:

- Link to the Git repository (GitHub/GitLab/Bitbucket)
- Evidence of version control and collaboration

12. Conclusion

Summarise:

- What the project achieved
- How the system addresses the problem
- Possible future improvements

Submission Format

- Submit the report in **PDF format**
- **Times New Roman**
- Font size: **12**
- Use clear headings and numbering
- **Include screenshots where relevant**
